Majid Manzarpour @majidmanzarpour Jul 5

real-time procedural dungeon generator in @threejs, built with Fable 5 👇

0:00 / 0:47

Jul 5, 2026 · 12:15 PM UTC

60 145 2,413 220,272

Majid Manzarpour @majidmanzarpour Jul 5

Open sourced by popular demand! Added multiple dungeon types and improved objects. There's a live demo link in the repo.

> 
>
> **Majid Manzarpour** @majidmanzarpour Jul 5
>
> open sourced my @threejs procedural dungeon generator, enjoy 🪄
> github.com/majidmanzarpour/t…

2 1 31 4,346

Erman Malak @ermanmalak Jul 5

Replying to @majidmanzarpour @threejs

this is dope. are you planning on making it explorable?

1 2 649

Majid Manzarpour @majidmanzarpour Jul 5

Yes, will incorporate it into gb.manzarpour.com

GRAVEBOUND - Co-op Gothic Survival ARPG

Band together against the endless dead. A free browser co-op gothic-fantasy survival ARPG - play with friends, no download.

gb.manzarpour.com

1 5 551

Very cool! Great work!

💬 1 🔁 ♥ 1 📊 708

Majid Manzarpour @majidmanzarpour Jul 5

Thanks! 🙏

💬 🔁 ♥ 📊 599

Jared Brown @jaredatjared Jul 5

Replying to @majidmanzarpour @threejs

Prompt worked perfectly. Huge thanks!

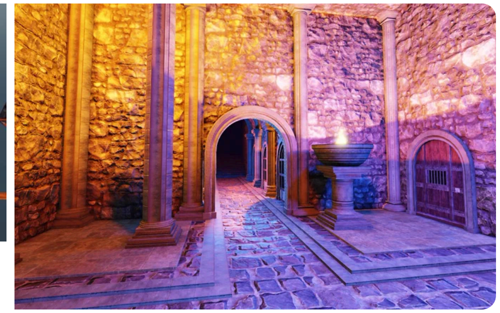
💬 1 🔁 ♥ 12 📊 2,165

Majid Manzarpour @majidmanzarpour Jul 5

Sweet! Just need to tweak the lighting to brighten it up

💬 🔁 ♥ 6 📊 1,729

FaryaBlender3D @FaryaBlender3D Jul 5 into fable to let fable make me different variations? And to what degree does Fable understand which modules fit together, does it try or do I have to give it clear instructions?

2 7 2,072

Majid Manzarpour @majidmanzarpour Jul 5

I think it could, if you provided the assets in your prompt or screenshots of them.

3 1,509

Len Seaside @LenSeaside Jul 5

Replying to @majidmanzarpour @threejs

Cant we just have yours and not waste credits?

2 5 907

Majid Manzarpour @majidmanzarpour Jul 5

I'll look into open sourcing, want to make some enhancements first

5 841

Brian @GrowthPact Jul 5

Replying to @majidmanzarpour @threejs

So much easier in threejs / html rather than Godot.

1 3 1,044

972

2bd @toobaddown Jul 5
Replying to @majidmanzarpour @threejs
What are you using for assets?
1 2 426

Majid Manzarpour @majidmanzarpour Jul 5
This is all procedural (pure code)
1 1 382

more replies

Eric @EricLevin77 Jul 5
Replying to @majidmanzarpour @threejs
dope!
1 2 817

Majid Manzarpour @majidmanzarpour Jul 5
🙏
799

spiralarchitect @spiral_archtct Jul 6
Replying to @majidmanzarpour @threejs
Absolutely sick, I want to see this as an AAA game now
1 1 486

Majid Manzarpour @majidmanzarpour Jul 6
It's being implemented in my game gb.manzarpour.com

GRAVEBOUND - Co-op Gothic Survival ARPG

Band together against the endless dead. A free browser co-op gothic-fantasy survival ARPG - play with friends, no download.

gb.manzarpour.com

1 3 471

more replies

Canzi @0xCanzi Jul 5

Replying to @majidmanzarpour @threejs

Can you post the git? please 😀

2 1 1,030

Majid Manzarpour @majidmanzarpour Jul 5

Posted a prompt to get you started

> **Majid Manzarpour** @majidmanzarpour Jul 5
>
> Replying to @threejs

PROMPT — Procedural Dungeon Generator for an Isometric Three.js ARPG

ROLE
You are a senior gameplay/graphics engineer specializing in procedural content generation and real-time WebGL. You write deterministic, allocation-conscious JavaScript and understand Three.js instancing, draw-call budgets, and forward-lighting costs at a deep level. You ship code that self-verifies.

CONTEXT
Target: an isometric, Diablo-style ARPG built in Three.js (r128+, WebGL2). Camera is orthographic — yaw 45°, pitch 35–40°. World scale: **1 unit = 1 tile**, Y-up, grid (x, y) maps to world (x, z). Generation runs on the main thread on level load. No frameworks, no build step: one self-contained module.

OBJECTIVE
Implement `generateDungeon(params) → Dungeon` (pure data, zero THREE imports) plus `buildDungeonScene(dungeon) → THREE .Group` (pure presentation). Dungeons must have:
- **Character** — irregular silhouettes, mixed room shapes, themed set-pieces, torch-lit atmosphere, a seeded name.
- **Depth** — branching with dead-end rewards, guaranteed loops (never a pure tree), secrets off the critical path, tension/release pacing.
- **Challenge** — a difficulty ramp keyed to graph distance from the entrance, elite gates on the critical path, boss at maximal depth.

NON-NEGOTIABLE CONSTRAINTS
1. **Determinism.** One integer seed reproduces the dungeon bit-for-bit. `Math.random`, `Date .now`, and iteration order over unordered Sets/Maps feeding logic are banned. Use mulberry32 (or xoshiro128**), threaded explicitly through every stage.
2. **Connectivity.** Flood fill from the entrance must reach 100% of floor cells. On failure, re-roll internally with a derived seed (max 5 attempts) —

10 draw calls via `InstancedMesh` — zero per-tile `Mesh` objects. ≤ 12 dynamic point lights, no shadow maps on point lights.
4. **Separation of concerns.** Generator returns typed arrays + POJOs only. Renderer consumes that data and owns all THREE objects, with a `dispose()` that frees everything on regenerate.

PIPELINE — implement as discrete, individually testable stages
1. **RNG.** mulberry32 with helpers: `float(a,b)`, `int(a,b)`, `pick(arr)`, `chance(p)`, `gaussian(mu,sigma)` (Box–Muller).
2. **Room scatter.** Spawn `roomCount × 1.4` candidates inside an ellipse (radius ∝ √roomCount so density stays constant). Archetype table — small 5–7 (45%), medium 8–12 (40%), large 13–18 (15%). Shape table — rectangle 60%, ellipse 22%, chamfered octagon 18%. Force ≥ 2 large rooms.
3. **Separation.** Iterative AABB push-apart with 2-cell padding until stable (cap 300 iterations), snap centers to integer grid, cull the smallest overflow down to `roomCount`.
4. **Connectivity graph.** Bowyer–Watson Delaunay over room centers → Prim MST as the skeleton → re-add non-MST Delaunay edges with probability `loopChance` (default 0.15), rejecting edges longer than 2.2× the mean MST edge. Loops are mandatory for ARPG flow: report the cyclomatic number (E − V + 1) and require ≥ 1 at defaults.
5. **Semantics before carving.** Boss = largest-area room. Entrance = degree-1 room maximizing graph distance from the boss. Critical path = BFS entrance→boss; remaining leaves → treasure (cap 4); 1–2 shrines mid-depth off-path; 1–2 elite arenas on the critical path at 55–85% depth; everything else combat. `difficulty = 0.15 + 0.85 × (depth / maxDepth)`, boss = 1.0.
6. **Corridor carve.** L-corridors center-to-center with seeded elbow direction (skip the elbow when spans overlap enough for a straight run). Width 3 on the critical path, 2 default, 1 permitted on treasure spurs. Stamp FLOOR; tag corridor cells for styling and doorway detection.
7. **Rasterize.** `Uint8Array` grid of {VOID, FLOOR, WALL}; WALL = any VOID cell with an 8-neighbor FLOOR (render only these — back rows stay void and vanish into fog). Doorway = corridor cell 4-adjacent to room floor. Compute a per-cell BFS distance field from the entrance (`Int16Array`, −1

neighbors are all floor, ≥ 2 cells from any doorway). Torches on floor-facing walls with min Chebyshev spacing 4. Debris density ∝ `decorDensity`, higher in low-difficulty rooms. Braziers ringing the boss arena; one chest per treasure room; shrine crystal; entrance portal ring. Enemy spawns: `round(area / 18 × (0.5 + difficulty))` for combat/elite, none in entrance/treasure/shrine, never on a prop or doorway cell.
9. **Presentation metadata.** Seeded dungeon name from syllable tables ("The Ashen Vaults of Vor'gul"), per-room tint, stats block `{rooms, edges, loops, criticalLength, floorTiles, wallTiles, props, genMs}`.

RENDERING SPEC
- One `InstancedMesh` per kind: floor, wall, pillar, torch bracket, flame, debris, chest, spawn marker, crystal.
- **Baked per-instance AO:** floor color = base × (1 − 0.09 × min(adjacentWalls$_8$, 4)) ± 5% value noise, blended ~18% toward the room tint; corridors darker and untinted. Wall height 2.0 ± 0.25 seeded jitter for a ruined silhouette.
- **Lighting:** dim blue hemisphere + faint directional for form; warm point lights (0xff8c3a, distance ≈ 9, decay 2) on a farthest-point-sampled subset of torches within budget, plus entrance/shrine/boss key lights. Per-frame flicker (intensity noise + flame scale jitter). `FogExp2` over a near-black clear color gives ortho depth falloff.
- **Materials:** Lambert for level geometry (vertex lighting reads pleasingly chunky at tile scale and is cheap under many lights); unlit/emissive for flames, crystals, markers.
- **Debug overlays (toggleable):** Delaunay (faint), MST (white), loop edges (cyan), critical path (red), difficulty heatmap recolor.

DATA CONTRACT
```ts
interface Dungeon {
  params: Params; name: string;
  W: number; H: number;
  grid: Uint8Array;          // VOID | FLOOR | WALL
  bfs: Int16Array;           // per-cell distance from entrance, −1 = non-floor
  rooms: Room[];             // { id, cx, cy, w, h, shape, type, depth, difficulty,
```

  doorways: Cell[]; corridorCells: Cell[];
  props: Prop[];   // { kind, x, y, rot, scale, roomId }
  spawns: Spawn[];   // { x, y, tier, roomId }
  stats: Stats;
}
```

## ACCEPTANCE TESTS — must run automatically and print results
- Flood-fill reachability = 100% of floor cells.
- Same seed + params ⇒ identical grid checksum across 3 consecutive runs.
- Boss depth ≥ 60% of max BFS depth; entrance degree = 1; entrance ≠ boss-adjacent.
- ≥ 3 leaf rooms at 40+ rooms; loop count = cyclomatic number = E − V + 1.
- No prop or spawn on a doorway, wall, or void cell; light count within budget.
- 60-room generation < 50 ms (report the measured value).

## TUNABLES (expose with defaults)
`seed`, `roomCount = 42`, `loopChance = 0.15`, `decorDensity = 0.6`, `theme = 'crypt'`

## ANTI-GOALS
Uniform room sizes; corridor spaghetti (> 2 elbows per link); boss reachable in < 60% of max depth; per-tile meshes; shadow-mapped point lights; hidden nondeterminism; a pure-tree layout with zero cycles.

## DELIVERABLE
A single self-contained HTML file: generator module + scene builder + minimal control panel (seed input, dice, sliders for roomCount / loopChance / decorDensity, overlay toggles, live stats, legend). Include an optional staged build animation gated behind `animateBuild` — rooms scatter→separate, graph resolves Delaunay→MST+loops, floors flood
```

1,014

Dev VINRATA @dev_vinrata Jul 5
Replying to @majidmanzarpour @threejs
Is it just a 2d image? Like can you rotate it ? Will check it out later , super cool
1 1 520

Majid Manzarpour @majidmanzarpour Jul 5
It's 3D, you could add rotation yes!
1 472

Ulf @stilfuchs Jul 5
Replying to @majidmanzarpour @threejs
Dungeon Keeper next ;)

1 2 697

Majid Manzarpour @majidmanzarpour Jul 5
Classic 😎
2 591

XCancel(donate)

| |
Gone this doesn't have to be iso yea. Could just make it full 3d?

1 1 253

Majid Manzarpour @majidmanzarpour Jul 5

Correct, it's actually 3D just with iso perspective. You can scale it up and adjust.

244

Studio Zamudio One @StudioZamudio Jul 5

Replying to @majidmanzarpour @threejs

I tried to do something like this a while ago. It was a mess trying to automate the floor-plan rules. I finally just got it working and was so burned out on it, I never opened that folder again.

1 1 812

Majid Manzarpour @majidmanzarpour Jul 5

Check out the prompt I shared in the replies

1 1 720

more replies

Load more